

HW 1: Introduction to xv6

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Task 0.

- <https://github.com/Escanor4323/xv6--riscv-labs>
- TA GitHub email: danielmarin350@gmail.com – this repo has been shared.

Task 1. Boot xv6 and explore utilities

- used macOS as my host operating system and ran xv6 directly in my terminal using the QEMU RISC-V emulator (no full VM used).

QEMU and the RISC-V toolchain were installed using:

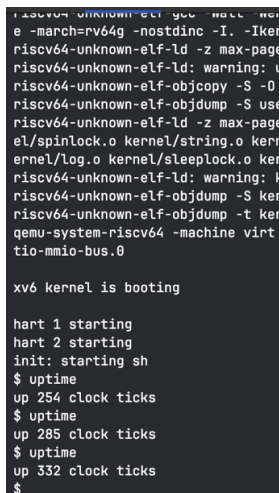


```
bash
1 brew install qemu riscv-gnu-toolchain
```

No full Linux VM was necessary because the xv6 Makefile builds everything for the RISC-V architecture and runs via `qemu-system-riscv64` in macOS.

Task 2. Implement the uptime utility

After building with `make qemu`, I booted into the xv6 shell and ran `ls`. The output looked like this:



```
riscv64-unknown-elf-gcc -march=rv64g -nostdinc -I. -Iker
riscv64-unknown-elf-ld -z max-page
riscv64-unknown-elf-ld: warning: u
riscv64-unknown-elf-objcopy -S -O
riscv64-unknown-elf-objdump -S use
riscv64-unknown-elf-ld -z max-page
el/spinlock.o kernel/string.o kern
ernel/log.o kernel/sleeplock.o ker
riscv64-unknown-elf-ld: warning: k
riscv64-unknown-elf-objdump -S ker
riscv64-unknown-elf-objdump -t ker
qemu-system-riscv64 -machine virt
tio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ uptime
up 254 clock ticks
$ uptime
up 285 clock ticks
$ uptime
up 332 clock ticks
$
```

Three Commands Explored

1. echo

- Description: Prints arguments to stdout.
- Code location: user/echo.c
- Example:

```
$ echo Hello for Operating systems
Hello for Operating systems
$
```

2. ls

- Description: Delays execution for N clock ticks.
- Code location: user/ls.c
- Example:

3. cat

- **Description:** Outputs the contents of a file.
- **Code location:** user/cat.c
- **Example:**

```
$ cat README
xv6 is a re-implementation of Dennis Ritchie's and Ken Thompson's Unix
Version 6 (v6).  xv6 loosely follows the structure and style of v6,
but is implemented for a modern RISC-V multiprocessor using ANSI C.

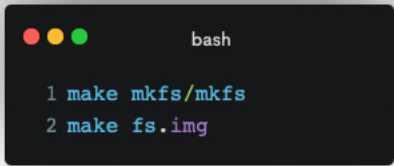
ACKNOWLEDGMENTS

xv6 is inspired by John Lions's Commentary on UNIX 6th Edition (Peer
to Peer Communications; ISBN: 1-57398-013-7; 1st edition (June 14,
2000)). See also https://pdos.csail.mit.edu/6.828/, which
provides pointers to on-line resources for v6.

The following people have made contributions: Russ Cox (context switching
locking), Cliff Frey (MP), Xiao Yu (MP), Nickolai Zeldovich, and Austin
Clements.
```

Difficulties Encountered

- Initially, make qemu failed because **QEMU** or the **wrong RISC-V toolchain** was installed (riscv64-linux-gnu instead of riscv64-unknown-elf).
- I fixed this by installing the correct toolchain via Homebrew (riscv-gnu-toolchain) and making sure riscv64-unknown-elf-gcc was on the PATH.
- Another issue was that fs.img failed to generate because mkfs didn't build. This was solved by running:



```
bash
1 make mkfs/mkfs
2 make fs.img
```

A terminal window with a dark background and three colored window control buttons (red, yellow, green) in the top-left corner. The title bar of the window says "bash". The terminal shows two lines of commands: "1 make mkfs/mkfs" and "2 make fs.img".